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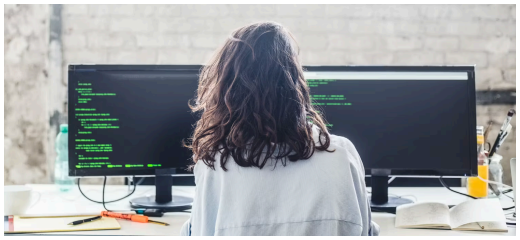
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What Is a Computer Engineer? 2026 Career Guide

Written by Coursera Staff • Updated on Dec 8, 2025
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Learn about the career of a computer engineer, including what they do, their earning potential, and how to become one.



Key takeaways

Computer engineers combine computer science and electrical engineering disciplines to design and build computer hardware and systems. A few things to know:

- Their primary role is to enhance the functionality and accessibility of devices and systems, such as medical devices and communication networks.
- Computer engineers often work for computing companies in the tech field or in industries like automotive, government, aerospace, and telecommunications.

- You can become a computer engineer by pursuing a degree, earning credentials through certification programs, and gaining experience in roles like computer systems analyst or software engineer.

Read on to learn more about the career of a computer engineer and how to become one. If you're interested in building the skills you need to transition into this industry, consider enrolling in the Mathematics for Engineers Specialization offered by the Hong Kong University of Science and Technology. In as little as three months, you could learn to analyze basic numerical techniques and write scientific computation programs in MATLAB.



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What do computer engineers do?

Computer engineers design, develop, and maintain computer components and systems, often specializing in a particular area of computer engineering, including hardware, software engineering, and networks. This field is evolving, with certain specializations growing with the expansion of technologies such as artificial intelligence and robotics.

Employing computer engineers allows companies and organizations to ensure they’re updated with the newest technologies, networks, and systems. Their processes typically involve designing, building, implementing, and

continually maintaining different computer components. Successful computer engineers utilize principles across numerous fields like computer science and mathematics.



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Computer engineer tasks and responsibilities

Your daily responsibilities as a computer engineer vary depending on factors such as the industry you work in and your job description. Here are some common tasks computer engineers perform:

- Designing, developing, and testing various computer hardware and software products
- Developing and maintaining information technology components such as networking systems and circuits
- Troubleshooting and testing system capabilities and analyzing results to make any necessary adjustments
- Collaborating with developers, clients, technical support, and others to ensure the best product quality and compatibility between hardware and software

- Researching advancements in relevant computer engineering methods and technology to improve innovation

What tools do computer engineers use?

Regarding engineering components, you can implement different tools to simplify the hardware and software development process. Here are some tools that computer engineers may use:

- GitHub provides a central location for sharing and collaborating on projects with other hardware and software engineers. It also has useful project management features used for performing code reviews.
- MATLAB is used to analyze and design systems with a matrix-based language.
- **Computer-aided design (CAD)** platforms are used by hardware engineers to visualize prototypes.

Job outlook and pay

As a computer engineer, your salary can vary depending on certain factors such as years of experience, education level, and industry. Here is a breakdown of average annual salaries in the US for different computer engineers, according to Glassdoor (as of October 2025):

- Hardware engineer: $148,000 [1]
- Software engineer: $161,000 [2]
- Network engineer: $123,000 [3]

Read more: Computer Engineer Salary: Your Guide

As the world continues to digitalize, the US Bureau of Labor Statistics (BLS) reports that computer engineers are in high demand with a faster-than-average job outlook compared to all other occupations. The following are projected job growth for the different types of computer engineers according to the BLS from 2024 to 2034:

- Computer hardware engineers: 7 percent [4]
- Network engineers: 12 percent [5]
- Software engineers: 15 percent [6]

How to become a computer engineer

To become a computer engineer, you'll need to meet both the education and skill requirements. Once obtaining those, consider earning certifications to ensure you're equipped with the latest information and skills in your field.

Read more: Is Computer Engineering Hard?

Do computer engineering jobs require a bachelor's degree?

Sixty-six percent of computer engineers have a bachelor's degree, 14 percent have a master's degree, and 11 percent have an associate degree [7]. Earning a degree from an Accreditation Board for Engineering and Technology (ABET) program is preferred by some employers and can be beneficial in ensuring you meet hiring standards set by the ABET. However, education requirements depend on the employer, industry, and company. Continuing your education to earn a master's degree is a great option for increasing your earning potential and career growth opportunities.

Is computer engineering a program of study?

Yes, computer engineering is a program of study available at many universities. Twenty-eight percent of computer engineers majored in computer engineering, 18 percent in computer science, 18 percent in electrical engineering, and five percent in business [7].

Read more: Computer Science vs. Computer Engineering: How the Roles Differ

Technical skills

Depending on the area of computer engineering you specialize in, certain skills are more relevant than others. Technical skills represent the practical aspects of your role, here are some valuable ones that computer engineers need:

- Computer programming
- Software development
- Cloud computing
- Operating systems
- Debugging
- Microprocessor design
- DevOps
- Signal processing

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Workplace skills

In addition to having strong technical skills, developing the following workplace skills can help you succeed in computer engineering. This type of skill contributes to how productive you are in a work environment, here are some to have:

- Creativity
- Communication
- Problem-solving
- Critical thinking
- Analytical thinking
- Teamwork

Read more: Is Computer Engineering Hard?

Certifications and other professional credentials

Earning a certification demonstrates your latest skills and knowledge in the field to your employer. Depending on the type of computer engineer you are, consider exploring the certifications that relate to your profession. Some jobs might require a specific certification, so it's a good idea to research some job listings of roles you're interested in before preparing for a certification exam.

Read more: Are IT Certifications Worth It? When to Get Certified in Your IT Career



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